

Grade Eight Science - Statics Problems to Start the Class - ANSWERS

Problem 1

Analyze the information given in the table below and indicate whether unknown "substance X," falls above or below the material given

Charging Situation	Is "substance X" below or above the given material on the triboelectric series
Rubbing cotton and "X" together gives "X" a positive charge	Below cotton (X gives negatives)
Rubbing wool and "X" together gives "X" a negative charge	Above wool (X takes negatives)
Rubbing polyester and "X" together gives "X" a positive charge	Below polyester (X gives negatives)
Rubbing silk and "X" together gives substance "X" a negative charge	Above silk (X takes negatives)

Problem 2

Outline how you can charge a neutral plastic ball to make it positive? A person wraps their hand around a positive plastic ball. **State** the purpose of doing this and **state** the charges will be left on the hand and the ball?

In order to make a neutral plastic ball positive, you can rub it with a neutral material that is above plastic on the Triboelectric series, such as polyester. Since polyester has a higher electron affinity than plastic, the plastic ball will give electrons to the polyester if they are rubbed together (Charging by friction). This will leave the plastic ball positive since it lost electrons.

Putting a neutral hand around the positive ball grounds the ball meaning that it should bring the ball back to neutral. Since the ball is charged positively, it sees the neutral hand as a chance to get back to neutral. When they come into contact the ball can take electrons from the hand to balance out its negatives and positives. This would leave the ball neutral and the hand positive since the hand gave electrons to the ball.

Problem 3

A student rubs two balloons against her hair. When the balloons are laid on the table they move away from each other. **Describe** the reason why the balloons move away from one another. Can either of these balloons be neutral if they are moving away from each other?

The balloon is made of rubber. On the Triboelectric series, rubber is above hair meaning that hair has a lower electron affinity than the rubber. When charging by friction occurs, electrons are transferred from the hair (lower EA) to the rubber balloon (higher EA). This makes each balloon negative.

Since the balloons are both negative, they will repel each other because like charges repel. This is why they move away from each other. There is no neutrality in repulsion; in order to repel both objects must have the same charge so they can't be neutral.

Problem 4

A positively charged metal ball (Ball X) is touched against a neutral metal ball (Ball Y). When they are separated what charge is left on each ball. **Describe** how you know.

When a Charged object comes in contact with a neutral object a Charged object uses this as an opportunity to get itself back to neutral.

Positive ball X will need to gain negative charges in order to become neutral Because it has to balance out the positive and negative charges.

When the positive ball comes in contact with neutral ballBallTakes electrons from neutral ball in order to make ball X neutral. This leaves ball Y positive because it gave negative electrons to ball X in order to neutralize it

Problem 5

A plastic rod is rubbed with a piece of silk. Afterward, the silk repels a ping pong ball. The plastic rod is brought near a styrofoam ball and it attracts the ball, but they do not touch. Use scientific knowledge to **outline** what is going on in this scenario.

Assuming that both the plastic rod and piece of silk are neutral at the start of the experiment, plastic has a higher electron affinity than silk. Therefore when these materials are rubbed together, an example of charging by friction, electrons will move from the silk to the plastic since silk has a lower electron affinity than plastic. This makes the plastic negative (because it gained electrons from the silk) and the silk positive (since it lost electrons to the plastic)

If the silk is now positively charged and it repels a ping-pong ball then the ping-pong ball must also be positively charged. There is no neutrality in repulsion so the objects must have the same charge in order to repel.

If the plastic rod is brought near the styrofoam ball and the styrofoam ball attracts the plastic rod but they do not touch, the styrofoam ball could either be neutral or positive. Positively charged objects will attract negatively charged objects. Charged objects can also be attracted to neutral objects causing neutral objects to polarize.

The charge on the styrofoam ball did not change because it never came in contact with the plastic rod, so it must either be neutral or positive; there is no way for us to tell.

Problem 6

An ebonite rod is rubbed with human hair. The rod attracts a styrofoam ball. The ball sticks onto the rod and then falls to the ground. Use scientific knowledge to **describe** what is going on in this scenario.

We will assume that both the ebonite rod, human hair, and styrofoam ball are neutral at the start of the experiment.

Since human hair is lower than Ebonite on the triboelectric series, when they are rubbed together the hair will put electrons on the ebonite rod making the rod negative and the hair positive.

The negative ebonite rod attracts the neutral styrofoam ball. This causes the charges on the styrofoam ball to polarize with the positive charges moving close to the rod and the negative charge is moving away from the rod.

When the neutral styrofoam ball comes into contact with the negative ebonite rod, the negative rod sees it as the chance to get back to neutral. Once they are in contact the rod gives negatives to the styrofoam ball making the ball negative.

The styrofoam ball could have fallen back down due to repulsion. Sometimes not enough negatives will move from the rod to the ball. This means that the ball becomes negative

because it gained electrons but the rod didn't lose enough electrons to neutralize itself so it also remains negative. This means that they will both remain negative causing the ball to fall off.